

Eric Solak

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EDUCATION

McMaster University | Hamilton, ON

2022 - May 2026

Bachelor of Engineering, Software Engineering

cGPA: 3.70/4

EXPERIENCE

Alloy | Toronto, ON

Jun 2026 - Present

Founder & Technical Lead

- Grew Alloy into a **solo consultancy generating over \$25k** in contracted revenue across automation and website engagements.
- Design and build production systems end to end: custom websites in React and TypeScript deployed on Cloudflare Workers, and fully/semi-autonomous AI automation pipelines tailored to each business's existing systems.
- **Delivered 10+ end-to-end automation and website projects** for clients across Ontario, working solo across sales, scoping, build, and delivery.
- Quoting custom website engagement up to \$11,000+, alongside a pipeline of smaller template and semi-custom builds.
- Built automation systems collectively **saving client businesses 100+ hours of manual work per month** across active engagements.
- Maintained a 1 to 3 week average delivery window per project while **handling all client-facing and technical work single-handedly**.
- Sustained an **80%+ client retention or referral rate** through direct, hands-on relationship management with every client.

CATTLElytics Inc. | Hamilton, ON

Sep 2025 - Apr 2026

Software Engineer

- **Led AI development** for Pocket AI, an AI-powered hands-free app for workers who can't stop to type: it listens to natural speech to create and manage tasks in real time. Delivered as a complete product to CATTLElytics, a platform managing **1 in 50 cows across North America**.
- **Directed a team of 5 engineers** through the full product development lifecycle in **8 months**, from requirements gathering and architecture design to production handoff, delivering a fully functional application now being integrated into CATTLElytics' platform.
- Integrated OpenAI Whisper for real-time audio transcription, validated through usability testing that achieved **98% system voice recognition accuracy** across 200 test commands under simulated barn noise, exceeding the 95% project target.
- Built the Flask backend powering voice-to-task workflows with REST APIs for authentication, task lifecycle management, with an **offline-first architecture**, ensuring reliable performance in low-connectivity farm environments.
- Designed and built mobile UI components in React Native (Expo), including task views, audio capture flows, and speech confirmation feedback; usability testing yielded a mean satisfaction score of 4.4/5.

PROJECTS

musicGPT · 45M-Parameter LLM from Scratch | Python, PyTorch

Jul 2026

- Built a **45M-parameter decoder-only transformer GPT entirely from scratch**, hand-implementing RoPE, causal self-attention, LayerNorm, and the training loop.
- Trained on a 0.92B-token pipeline with a hand-rolled markup stripper and 3x upweighting of music-domain content; **reduced val loss from 10.9 (uniform baseline) to 3.26** with no overfitting across 14,000 steps.
- Wrote a pytest suite for silent failure modes (attention equivalence, causality, RoPE geometry) that caught a **weight-tying logit leak** and a **vocab-padding crash** invisible to standard training metrics.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, SQL, C

Frameworks & Libraries: PyTorch, FastAPI, Flask, React Native, NumPy, Pandas, scikit-learn

Tools: Docker, AWS EC2, Render, Claude Code, GitHub Actions, JUnit, SQLite

INTERESTS

Self-taught musician (piano & guitar, 10 years) · chess · weightlifting · running